



फलसु

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④ AI: Artificial Intelligence has many benefits but addressing it has some risks like create unemployment, job taking jobs and it is also humanly dependence on machine.

⑤ Big data: big data compared a large amount of data due to this it is difficult to collect and manage the whole data.

⑥ IoT: like the smart IoT has many benefits but it has a problem like security, privacy, surveillance. - problem.
- it expose the user information to various kind of attackers.

⑦ Generally, these technologies have both benefit and cost (risk)

5. What are the challenges (disadvantages) of IoT? (List out at least four) (2 pts)

- 1- As the number of connected devices increases and more information shared via the devices, the potential of hacker control also increase.
- 2- If there is a bug/vulnerability in the system, every connected device will become corrupted/damaged.
- 3- Since there is no international standard of compatibility, the devices communicate each other is difficult.
- 4- Managing and collecting data from large massive devices become challenged.

Part-III: Answer the following Questions

1. Write at least three professional responsibilities in Emerging technologies? (3pts.)
 1- Responsible use of technology: Professionals should use technology responsibly and ethically, ensuring that their actions do not harm others or society.
 2- Continuous learning: Professionals should stay updated with the latest trends and advancements in emerging technologies to remain relevant in their field.
 3- Transparency: Professionals should be transparent about their use of technology and the data they collect, ensuring that users are aware of how their information is being used.
2. List out and briefly explain the three main blocks of Augmented Reality System (ARS) architecture (4pts.)
 1- Input devices: These are the devices that provide input to the AR system, such as sensors, cameras, and tracking devices.
 2- Processing unit: This is the central unit that processes the input data and generates the AR content.
 3- Output devices: These are the devices that display the AR content to the user, such as head-mounted displays, smart glasses, and mobile devices.
 4- Content creation: This is the process of creating the AR content, which involves designing 3D models and animations that can be overlaid on the real world.
3. Write at least three application areas of IoT, and discuss them briefly? (3pts.)
 1- Smart homes: IoT can be used to create smart homes where appliances, lighting, and security systems can be controlled remotely via a smartphone or voice assistant.
 2- Healthcare: IoT can be used in healthcare for remote patient monitoring, drug delivery, and medical research.
 3- Industrial automation: IoT can be used in industrial settings for automation, predictive maintenance, and supply chain management.
4. New and emerging technologies (such as AI, Big Data, IoT) pose significant opportunities for businesses if we utilize them well and understand their true value early on. They also pose risks and questions not only to business but to society as a whole. Write the risks of those emerging technologies and briefly explain them? (4 pts.)
 1- Job displacement: Automation and AI may lead to the displacement of jobs, particularly in manufacturing and routine tasks.
 2- Privacy concerns: The collection and analysis of large amounts of data can lead to privacy violations and the potential for surveillance.
 3- Algorithmic bias: AI systems can perpetuate and even amplify existing biases and discrimination if they are trained on biased data.
 4- Security risks: The interconnected nature of IoT devices and systems can create vulnerabilities that can be exploited by cybercriminals.

14) Which one of the following statements correctly describes Mixed Reality(MR)?

- A. The synthetic content and the real-world content are able to react to each other in real-time.
- B. MR is a digital overlay that allows interactive virtual elements to integrate and interact with the real-world environment.
- C. It is the ability to have one foot in the real world, and the other in an imaginary place.
- D. All

15) What is a sensor?

- A. Anything that stores data.
- B. Something that provides products and services to customers through an internet connection.
- C. The foundation of the digitized world.
- D. A device that detects or measures an event or physical property.

16) Which one of the following describes the role of sensors in IOT architecture?

- A. Capture the physical position and location of the device.
- B. Measure the angular movement of a device.
- C. Help the devices to take autonomous decisions according to the changes of a device's peripheral.
- D. All of the above

17) AR provides the following benefits to patients and healthcare workers, except one:

- A. Increase the risks associated with minimally invasive surgery.
- B. Better informed decisions about the right treatment and illness prevention.
- C. Medical training and education.
- D. Assistance in medical procedures and routine tasks.

18) Which of the following IoT layer is a user-centric layer that executes various tasks for the users?

- A. Data Processing layer
- B. Network Layer
- C. Application Layer
- D. Sensing Layer

19) What is the goal of Augmented Reality?

- A. to improve and enhance our perception of the surroundings
- B. to connect physical objects to the Internet
- C. to providing trust and quality of information
- D. None

20) _____ exploring appropriate and ethical behaviors related to online environments and digital media?

- A. Cyber attack
- B. Cyber safety
- C. Cyber ethics
- D. Cyber security

- 5) Which one of the following is responsible for measuring the change in movement and the orientation of IOT devices?
 A. Actuator
 B. Magnetic sensor
 C. Motion sensor
 D. GPS sensor
- 6) Which of the following is an example of Augmented Reality experiences?
 A. Oculus Rift
 B. Pokémon Go
 C. HoloLens
 D. HTC Vive
- 7) Which one of the following alternatives is correct about digital privacy principle?
 A. The information collected and used must be provided in a clear manner.
 B. Information collected will not be used for any other purpose unless authorized or mandated by law.
 C. Information collected will be maintained by ensuring the interests of the individuals and businesses are protected.
 D. All of the above
- 8) Which of the following is not the benefit of IoT in a Smart City?
 A. Intrusion detection systems
 B. Lighting
 C. Structural Health
 D. Waste Management
- 9) Which one of the following is false about IoT?
 A. IoT consists of any device with an on/off switch connected to the Internet.
 B. IoT consists of a gigantic network of internet-connected things and devices.
 C. IoT loses its distinction without sensors.
 D. IoT devices requires human help to transfer data over a network
- 10) Both Augmented Reality (AR) and Mixed Reality (MR) process digital content and real-world elements. Which one of the following statements is True?
 A. MR overlays digital information on real-world elements.
 B. AR is the merging of real and virtual worlds to produce new environments.
 C. MR allows you to see and immerse yourself in the digital world.
 D. AR keeps the real world central but enhances it with other digital details.
- 11) In which of the following case individual privacy is breached?
 A. The company or business has made personally identifiable information public.
 B. The client messages and records being compromised and leaked.
 C. Data concerning a personal preference is leaked.
 D. The user interrupted by receiving unwanted ads and emails.
- 12) Which one of the following is not labeled as the advantage of IOT?
 A. Technology Optimization
 B. Waste Reduction
 C. Compliance
 D. Enhanced data collocation
- 13) Which IoT features transform a standard passive network of devices into an active system capable of real-world integration.
 A. Connectivity
 B. Sensors
 C. Active engagement
 D. AI

BAHIR DAR UNIVERSITY
BAHIR DAR INSTITUTE OF TECHNOLOGY
FACULTY OF COMPUTING

Introduction to Emerging Technologies Mid Exam

Total mark: 30% | Time allowed: 60 Minutes | Date: August 11, 2021 | Target group: 1st Year Students

General Instruction

- Switch off your phone
- Write answers precisely and concisely.
- No use of calculator is permitted.

Name [Redacted] ID 1310020 Section 6A

Part I. Write true if the statement is correct and false if the statement is incorrect (5 pts.)

- ~~False~~ 1. NoSQL technologies are designed to store data in a row and column format.
- ~~False~~ 2. Currently available AI systems cannot beat the imaginative and creative thinking power of human beings.
- ~~False~~ 3. The primary goal of HCI is to improve the interaction between computers to computers.
- ~~False~~ 4. A much bigger percentage of all the data in our world is semi-structured data.
- ~~False~~ 5. HDFS is a centralized file system in Hadoop ecosystem.

Part II. Choose the best answer from the given alternatives (15 pts)

- ~~C~~ 1. Which of the following is the goal of data curation in the Big Data Value Chain?
- A. Persistence of data in a scalable way
 - B. Conducting data-driven business activities
 - C. Ensuring quality of data
 - D. Highlighting relevant data
- ~~C~~ 2. Which of the following is an example of unstructured data?
- A. Spreadsheet
 - B. XML
 - C. NoSQL databases
 - D. SQL databases
- ~~A~~ 3. Which of the following device is responsible for executing software instructions in digital electronic systems?
- A. Microprocessor
 - B. Logic Device
 - C. Storage device
 - D. Network device
- ~~A~~ 4. Which of the following component of AI systems is responsible for taking action?
- A. Interacting layer
 - B. Interpreting layer
 - C. Sensing layer
 - D. Perception layer
- ~~D~~ 5. The type of AI machines which only focus on current scenarios and react on it as per possible best action.
- A. Self-Awareness
 - B. Limited memory
 - C. Theory of mind
 - D. Reactive Machine

6. A highly increasing amount of data, which cannot be stored, processed, and analyzed using the Relational database management systems.
 A. Data science
 B. Big data
 C. Hadoop
 D. NoSQL
7. Which one of the following, can't be considered as the key features of Hadoop?
 A. Reliability
 B. Low Scalability
 C. Flexibility
 D. Economical
8. Which one of the following is the influencer of AI, which enables software components to communicate with each other?
 A. Big data
 B. Cloud computing
 C. Data science
 D. API
9. Which Industrial revolution era is characterized by electricity and mass production?
 A. Digital
 B. Technological
 C. Mechanical
 D. AI
10. Which one of the following can be the motivating factor for the development of IR 4.0?
 A. Artificial Intelligence
 B. Big data analytics
 C. Cloud computing
 D. All of the above
11. Which of the following is some sort of hardware architecture or software framework that allows the software to run?
 A. Hadoop file system
 B. Platform
 C. API
 D. Chat bot
12. The level of AI that is not available today.
 A. Context awareness and Retention
 B. Rule based systems
 C. Domain specific expertise
 D. Self-aware systems
13. The strength of narrow AI is:
 A. It can exhibit the capabilities of the human brain.
 B. It is able to capture scientific creativity, general wisdom, and social skills.
 C. It focuses on a specific task which enables to do tasks well.
 D. It is much smarter than the best human brains practically in every field.
14. Which of the following Hadoop component is used for a distributed data storage unit?
 A. HDFS
 B. MapReduce
 C. ZooKeeper
 D. Hadoop YARN
15. The field of artificial neural networks with several hidden layers is _____.
 A. Machine learning
 B. NLP
 C. Deep learning
 D. Cloud computing

Part III. Discuss and explain each question carefully

1. List at least four disadvantages of AI? (2 pts.)

✓ high cost.
 ✓ no original creativity.
 ✓ increasing dependence on machine.
 ✓ no feelings and emotion.
 ✓ unemployment
 ✓ can not think outside the box.

2. List and discuss at least three characteristics of big data. (2 pts.)

- 1- Volume :- large amount of data.
- 2- Velocity :- data at live stream/motion.
- 3- Variety :- consists of different types of data.
- 4- Veracity :- how the data can trust/accurate?
→ data in doubt

3. What are the goals of artificial intelligence (list at least 4)? (2 pts.)

- ✓ replicate human intelligence.
- ✓ solve knowledge-intensive tasks.
- ✓ intelligent connection of perception and action.
- ✓ create expertize system exhibit/done specific task.
- ✓ build a machine in order to do specific work.

4. List four disciplines contributing to Human-Computer Interaction (HCI) (2 pts.)

- ✓ computer science.
- ✓ cognitive psychology.
- ✓ engineering and design.
- ✓ language.
- ✓ artificial intelligence.

5. Is it possible to map the way of human thinking to artificial intelligence components? If your answer is Yes, how? (2 pts.)

Yes - It is possible to map the way of human thinking to artificial intelligence components by the following stages.

Stage 1 - observing and collecting the input of data/information from the surrounding env. + put in to the brain.

Stage 2 - interpret and evaluate the input of data coming from the surrounding env. +

Stage 3 - finally make decisions / take action.

So, it is possible to map the way of human thinking to AI components.

41.5
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BAHIR DAR UNIVERSITY
BAHIR DAR INSTITUTE OF TECHNOLOGY
FACULTY OF COMPUTING
Introduction to Emerging Technologies Final Exam
Total mark: 50% | Time allowed: 1:30 Hrs. | Date: Sep. 21, 2021 |
Target group: 1st Year Students

General Instruction

- Switch off your phone
- Write an appropriate answer for each question precisely and concisely.
- Attempting to copy from other students is prohibited.
- Put your answer on the space provided only

Name: _____

Part-I: Write true if the statement is correct and write false if the statement is incorrect (5 pts)

- 1) The content in Augmented Reality (AR) is 100% digital and can be enjoyed in a full immersive environment. 26.5
- 2) As with many technological advances, the Internet has been with negative aspects, that can serve to promote or restrict human rights.
- 3) With Technology moving at a fast pace, it is always been a challenge for security.
- 4) It is probably not contentious to say it would be desirable to have some ethical input at the earlier stages of technology design and development.
- 5) The communications between connected devices in IoT are guaranteed absolute security from any emerging threats.

Part-II: Choice Choose the best answer from the given alternatives (30 pts.)

- 1) A type of technology in which users are visually isolated from the real environment?
- A. Virtual Reality
B. Mixed Reality
C. Augmented Reality
D. Augmented Virtuality
- 2) Which of the following IoT layer is a user-centric layer that executes various tasks for the users?
- A. Data Processing layer
B. Network Layer
C. Application Layer
D. Sensing Layer
- 3) Which definition best fits augmented reality?
- A. Technology that turns physical objects into digital objects
B. Technology that overlays digital information on top of real-world items
C. Technology that completely immerses users in a new digital environment
D. Technology that can achieve a human-level understanding of images
- 4) Abebe sends a secured message for 'Alemitu' digitally, and 'Worku' access this message without their permission. Which privacy is compromised?
- A. Information privacy
B. Communication privacy
C. Individual privacy
D. None

- Smo - den -
- PS -
- con - cam -
- how - again -

Part I: Write true if the statement is correct or False if not. (1 Point each)

- ☐ 1. Virtual reality uses an existing environment and overlays new information on top of it.
- ☐ 2. Self-driving cars are one of the best examples of Limited Memory systems.
- ☐ 3. In the architecture of IOT, the application Layer is to identify any phenomena in the devices' peripheral and obtain data from the real world.
- ☐ 4. Cloud computing is a general term that describes the delivery of on-demand services through internet.
- ☐ 5. In domain-specific expertise levels of AI, the machine is trained on the knowledge and experience of the best humans.
6. Communications can be intercepted or delivered to other recipients without the sender's knowledge in a multitude of ways.
- ☐ 7. Unlike humans, artificial intelligence machines aren't creative and can't imagine some new ideas.

Part II: Choice the best answer from the alternatives. (1.5 Point each)

1. Which one of the following is not true about artificial Intelligence?
- A. It is about the copy of something natural
- ☒ B. It exists when a human can have machine-based skills ✓
- C. it is the creation of intelligent machines which can behave like a human ✓
- D. It deals with the area of developing computing systems that are capable of performing tasks that humans are very good at. ✓
2. Which one of the following is not the advantage of AI?
- A. High Accuracy with fewer errors
- ☒ C. Increase dependence on machines
- B. High reliability
- D. Digital Assistant
3. Which augment reality system architecture was designed for displaying the result?
- A. Processing Unit
- B. Infrastructure Tracker Unit
- ☒ C. Visual Unit
- D. All
4. Identify wrong statement about Augmented Reality
- A. It uses the existing environment and overlays new information on top of it ✓
- ☒ B. It is fully immersive.
- C. It allows users to overlay computer generated virtual content in the real world object
- D. It adds digital elements to a live view often by using the camera.
5. Which one is incorrect as the number of connected devices increases in IoT?
- A. The likely that a hacker could steal confidential information also increases ✓
- ☒ B. Information sharing along with connected IoT devices also increases.
- C. It's difficult for IoT devices to communicate with each other.
- D. Collecting and managing the data IoT devices will be challenging.

installed chess application in your mobile/ computer and playing chess game. This is an example of

- A. Internet of things
- ☒ B. Artificial intelligence

- C. Data science
- D. Augmented reality

7. Which technology is used to implement smart home automation?

A. Data science

C. Augmented reality

☒ B. Internet of things

D. Artificial intelligence

8. Which types of AI machines can store past experiences or some data for a short period of time?

A. Theory of Mind

C. Reactive Machines

B. Self-Awareness

☒ D. Limited Memory

9. Which of the following is influencer of Artificial Intelligence (AI)?

A. Big data and emergence of data science

B. Advancements in computer processing speed and new chip architectures

C. Cloud computing and APIs

☒ D. All

10. The complexity of IoT provides substantial personal data in extreme detail without the user's active participation. This IoT challenge is known as _____.

A. Security

☒ B. Privacy

C. Complexity

D. Compliance

11. _____ is the information acquired through experience.

A. Experience

B. Intelligence

☒ C. Knowledge

D. Data

12. Which level of artificial intelligence implies "What should I always do?"

A. Perception

C. Automation

B. Prediction

D. Prevention

Part III: Fill the Blank space (1 Point each)

1. Deep Learning is networks that extract features from the data in hierarchical fashion with several hidden layers.
2. In the architecture of IOT App layer is a user-centric layer that executes various tasks like the user.
3. HMD are currently the dominant display technology in the AR field.
4. APIs enable software components to communicate with each other.
5. In the AI System, Agent is anything that can perceive its environment through sensors and acts upon that environment through effectors.

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5. In the AI System, Agent is anything that can perceive its environment through sensors and acts upon that environment through effectors.

5. Which one of the followings is among high profiled examples of AI?
 A. Alexa
 B. Siri
 C. Pokman Go
 D. All
6. Which one of the following is not the result of IoT Based Smart Farming application?
 A. Animal Farming/Tracking
 B. Offspring Care
 C. Field Monitoring
 D. Waste Management
7. Which of the followings Technologies are influencing AI?
 A. Big data
 B. Cloud computing and APIs
 C. The emergence of data science
 D. All
8. Decision making in human thinking refers _____ in AI system.
 A. Sensing
 B. Interpretation
 C. Interaction
 D. all
9. Which one of the following is among currently emerging technology?
 A. Artificial Intelligence
 B. Cloud Computing
 C. Block-chain
 D. Augmented and Virtual Reality
 E. All

10. Which of the following is an AI application?

A. Spam Filters

B. Smart Email Categorization

C. Online shopping

D. all

11. Which of the following describes the networking of smart objects?

A. AI

B. IOT

C. Data science

D. A and C

12. What changes are observed as we go from the first level to the 7th level of AI?

A. An increase in intelligence capabilities

B. An increase in the realization of intelligence

C. An increase in self-awareness of AI systems

D. An increase in problem-solving capabilities

E. All are correct

13. Which of the following is true about IOT drawbacks?

A. If an error occurs in the system, it may affect the whole connected devices

B. Vulnerability to the security traits.

C. There are no adequate international standard set for its compatibility

D. As heterogeneous devices are connected, complexity is difficult to manage

E. All of the above

14. The Internet of Things operates at _____ scale.

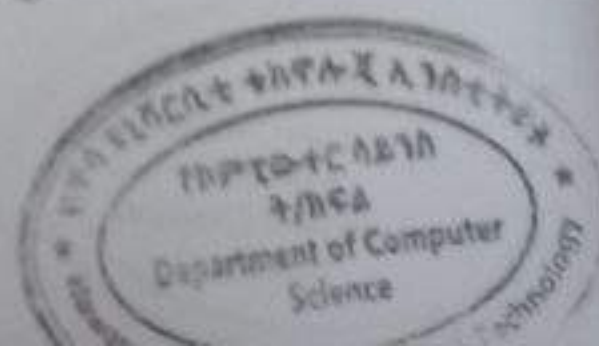
A. Machine

B. Human

C. Device

D. Sensor

15. Which stage of AI exhibit specific expertise knowledge though massive volume of information that will help us for decision making?



- A. Domain-Specific Expertise
 - B. Reasoning Machines
 - C. Self-Aware Systems / Artificial General
 - D. Artificial Super Intelligence (ASI)
16. Which of the following is drawback of AI?
- A. AI system can't think out of the box
 - B. It is free from feeling and emotions
 - C. It increases human dependence on machines
 - D. The AI system does not exhibit original creativity
 - E. All of the above
17. An AI system that has the ability to think, puzzle, make judgments, plan, learn, communication by its own can be categorized to which class of AI.
- A. Narrow AI
 - B. General AI
 - C. Super AI
 - D. Weak AI
 - E. A and D
18. Which of the following is not an application of IoT?
- A. Smart Devices/ Cities
 - B. Smart Agriculture/Connected Cars
 - C. Home Automation and Energy management (smart home)
 - D. All of the above.
 - E. None of the above
19. Which of the following is an application of Virtual Reality (VR)?
- A. Used for gaming, 3D cinema and in theme parks
 - B. Used for surgery training, exposure therapy for people with phobia or anxiety disorder
 - C. Used for astronaut training, flight training, military training, etc.
 - D. All of the above
20. Allowing Medical Surgeons to look at patient while they see an overlay of an x-ray is an example of _____.
- A. Virtual Reality
 - B. Augmented Reality
 - C. Mixed Reality
 - D. All of the above
21. Which one of the following is property of AI machine with domain-specific expertise?
- A. It has a sense of beliefs, intentions, knowledge, and how its own logic works
 - B. It goes beyond the capability of humans in a given domain it's designed for
 - C. It has human-like intelligence
 - D. It outsmarts the most intelligent humans in every domain
22. A collection of objects that can sense, communicate and share information, all interconnected over public or private networks?
- A. Network
 - B. Internet of Things
 - C. Cloud computing
 - D. API
23. Which one of the following is false about IoT?



Part III: Matching;

Find a match for the Item in column A from column B and write on the space provided (1 pt. each)

<u>A</u>	<u>B</u>
1. IoT System	A. Can think, learn, reason and act
2. Agent and Environment	B. Capability of AI
3. Theory of mind	C. IoT
4. To acquire information	D. Real world and objects with virtual contents
5. Automate Human Life	E. Consumer
6. Effectors	F. Act up on the environment
7. KAA	G. Functionality of AI
8. Intelligent Agent	H. IoT+ Network & Devices
9. IoT Device	I. AI System
10. Augmented reality	J. AI Sensing layer
	K. IoT Platform

Part IV: Fill in the blank;

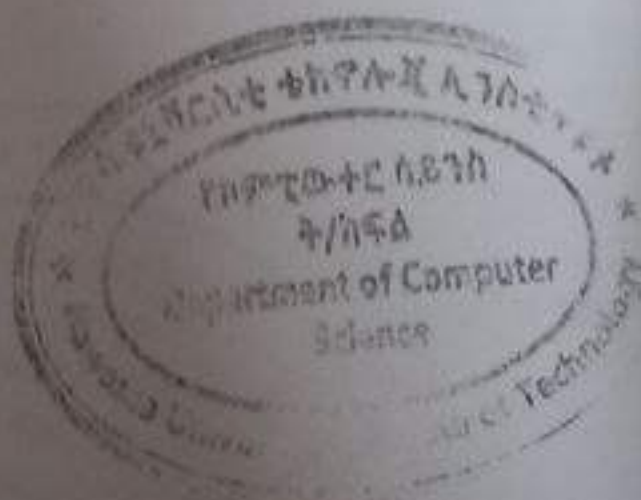
Write the answer for the following questions on the space provided (1 pt. each)

- _____ is the science of making machines imitate human thinking and behavior.
- _____ an emerging technology that aims to automate human beings' life.
- _____ is fully artificial environment, which tricks your mind into thinking you're in a different environment or world apart from the real world.
- An intermediary between IoT devices and applications is called _____.
- _____ consists of any device with an on/off switch connected to the Internet.
- _____ allow users to achieve deeper automation, analysis, & integration within a system in accurate manner.

Part V: Explanation /short Answer.

Write your answer on the space provided (Total 10 pts).

- List and discuss about IoT Architecture layers. (2 points)
- List and describe at least three Challenges of IoT system. (3 points)
- Explain the main influencers of artificial intelligence? (1 pt.)
- Explain the main advantage and disadvantage of IoT. (2 pts.)
- What is the difference between augmented reality and virtual reality? (2 pts.)



Part I: True or False:

Write true if the statement is correct and false otherwise on the space provide (1 pt. each)

1. One of the goals of AI is building a machine which can replicate human brain.
2. Narrow (Weak) types of AI learns like a human and can solve general problems as the human brain does.
3. IOT is a network of devices that can sense, accumulate and transfer data over the internet with a human intervention.
4. Augmented reality is fully immersive, which tricks your senses into thinking you're in a different environment apart from the real world.
5. Augmented reality allows interaction of real and virtual contents with each other in real time.

Part II: Multiple Choice:

Choose the best answer and write the letter of your choice on the answer sheet provided (1 pt. each).

1. Which one of the following is advantage of AI?
A. Digital Assistant
B. High accuracy with fewer errors.
C. Useful for risky areas
D. High speed.
E. All
2. What are smart devices in IoT platform?
A. Devices with sensing capability
B. Devices with computing capability
C. Devices with communicating capability
D. All of the above
3. In the architecture of Augmented reality, the Infrastructure Tracker Unit is responsible for _____.
A. Collecting data from the real world.
B. Overlaying or mixing the computer-generated content to the real-world view.
C. Creating a fully immersive environment.
D. Controlling the orientation and positioning of the virtual content.
4. Which one of the following statements is wrong about Augmented and Virtual Reality?
A. Contents in VR are 100% digital whereas not in AR.
B. VR is fully immersive whereas AR is not.
C. Contents interact in AR but not in VR.
D. User interacts with contents in AR but not in VR.



- A. Sensors are not required for establishing an IoT platform
 - B. IoT nodes or devices have AI capabilities
 - C. Connectivity of nodes or devices is always available
 - D. None of the above
24. Which of the following **feature is possessed** by a smartphone that supports auto-brightness?
- A. Motion sensor
 - B. Light sensor
 - C. Position sensor
 - D. Temperature sensor
25. A form of emerging technology that allows users to overlay computer generated content in the real world is called _____?
- A. Virtual reality
 - B. Augmented reality
 - C. Mixed reality
 - D. All
26. Which one of the following is considered as a limitation of HMDs?
- A. High price
 - B. Low mobility
 - C. User health concerns
 - D. All
27. Which of the following is/are benefits of using augmented reality in education?
- A. Students retain more knowledge for a long time
 - B. Help teachers attract the attention of their students and make lessons more effective
 - C. Help students develop their critical thinking skills
 - D. All
28. One of the advantages of IOT is to provide real-time information in order to make resource management more effective. This is generally referred to as:
- A. Technology Optimization
 - B. Improved Customer Engagement
 - C. Reduced Waste
 - D. All
29. Which layer of an AI system is mapped to acquiring information in human?
- A. Interacting layer
 - B. Interpretation layer
 - C. Sensing layer
 - D. Make decision
30. Which one of the following is used a virtual reality device?
- A. HTC Vive
 - B. Oculus Rift
 - C. Google cardboard
 - D. Gaming console
 - E. All of the above

